

Model review and revisions

JMP progress, September 17–October 2, 2026

Tommaso De Santo

I reviewed the model and calibration, including preferences, credit, demographic accounting, and the empirical targets.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Tested equivalence scales and changing housing shares.
- Rebuilt the first-birth housing evidence and national input measures.
- Revised adult entry and tested how purchase timing affects the calibration.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[\frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here h is rooms, m is children at home, and $\tau \in \{R, O\}$ denotes tenure. Parents require $h > h_P$; χ raises housing services for owners. The equivalence scale $e(m)$ increases family consumption needs. Child benefits are zero at $m = 0$; ψ sets their level and γ controls diminishing marginal benefits.

Demographic and market equilibrium

Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines H_0 .
- A balanced payroll tax finances equal retiree pensions.

Housing around the first birth

We found a mistake in the earlier regression and rebuilt the PSID event study using interview dates and household status before birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Credit timing

With $\phi = 0.8$, a purchase requires a 20% down payment. Purchase timing determines which resources count toward it. The soft rule includes current-period income in the purchase screen; the other rules use:

$$\text{Hard: } A \geq (1 - \phi)Q,$$

$$\text{Quarter-saving: } A + \frac{1}{4} [b' - (A - Q)] \geq (1 - \phi)Q.$$

Here A is financial wealth plus net proceeds from selling the previous home, Q is the cost of the chosen home, and b' is the end-of-period financial balance. Net saving is $b' - (A - Q)$.

For a home costing 100 with 10 available at closing, the hard rule blocks the purchase. The quarter-saving rule meets the 20-unit equity requirement if net saving is 40:

$$10 + 0.25 \times 40 = 20.$$

This approximates saving before purchase within a four-year period. Under quarter-saving, income helps only if saved. Hard and quarter-saving also impose an end-of-period net-estate floor for buyers; this is an additional change from soft.

More credit raises ownership but not births

Quarter-saving rule; financed share 80% vs. 100%; house price and parameters held fixed

	80% financing	100% financing
Ownership, ages 30–55 (%)	64.6	74.0
First-birth flow (relative change)		–0.5%
Completed fertility	2.100	2.090
<i>Households entering at ages 18–21 (same households in both)</i>		
Ownership (%)	10.5	31.4
First-birth probability (%)	25.5	25.8

Recalibrating under the hard and quarter-saving rules gives the same sign: first births fall by about 0.5%.

Why the response is small: what we have checked

- Rent per room equals the owner's cost per room, and χ does not depend on children. Credit makes buying easier but does not make a child's space cheaper.
- 60% of renting parents are at the six-room cap. Without the cap the median would rent 7.3 rooms. Holding saving fixed, the cap costs them about 2% of consumption in that period.
- The parent requirement h_P is the same for one, two or three children.

These are candidate explanations. We have not yet shown how much of the small response each one accounts for.

What I want to try next

- Selling costs: check whether the cost of trading up after a birth is what keeps credit from reaching fertility, and whether the 6% seller-paid cost is the right convention.
- Small owner homes: young buyers may lock into homes too small for a family.
- A housing requirement that grows with the number of children.
- Recalibrate before drawing conclusions from any of these.

Parameters and target moments

2007 stationary economy: ten parameters are estimated jointly from ten moments.

Parameter	Economic interpretation	Target moments
β	Patience	Wealth / earnings (6.927)
θ_0	Bequest weight	Annual bequests / wealth (0.729%)*
χ	Owner housing premium	Ownership, ages 30–55 (67.626%); mean rooms (5.729)*
κ_T	Tenure tastes	Ownership, ages 30–55; recent-parent gap (12.761 pp)
h_P	Basic space for parents	First-birth rooms (1.465)*; mean rooms
ξ	First-birth cost	Childlessness, ages 40–44 (19.828%); children at 25 (0.810)
κ_1	First-birth taste dispersion	Mean first-birth age (25.976); children at 25
κ_C	Later-birth taste dispersion	One child among mothers (21.366%); children at 25
ψ	Child-benefit level	Childlessness; children at 25
γ	Child-benefit curvature	One child among mothers (21.366%); children at 25

*Recomputed: mean rooms (AHS), bequests (SCF), and first-birth rooms (PSID). Also updated pension/earnings, depreciation and property tax. Parameters jointly affect the moments; H_0 follows from housing clearing and θ_1 is fixed externally.

Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

Moment	Target	Soft*	Hard	Quarter-saving
Completed fertility	2.100	2.100	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.222	20.413	20.546
One child among mothers, ages 40–44 (%)	21.366	21.264	20.907	20.844
Mean first-birth age (years)	25.976	26.032	26.007	26.057
Wealth / earnings	6.927	6.551	6.610	6.645
Annual child bequests / wealth (%)	0.729	0.685	0.684	0.685
Mean occupied rooms	5.729	5.957	6.241	6.131
Ownership, ages 30–55 (%)	67.626	66.527	60.878	64.572
First-birth rooms difference	1.465	1.324	1.063	1.167
Recent-parent ownership difference (pp)	12.761	11.825	10.125	11.683
Children at age 25 (capped at three)	0.810	0.528	0.524	0.521
Validation moments	Target	Soft*	Hard	Quarter-saving
First births at age 30+ (%)	24.928	23.448	23.078	23.395
Older wealth/income p90/p50	3.516	2.968	2.861	2.937
Family rooms difference	0.385	0.304	0.321	0.316

Soft*: best fit before the search stopped; final numerical check pending.

Parameter estimates

Estimates under the three purchase rules; H_0 follows from housing clearing.

Parameter	Economic interpretation	Soft*	Hard	Quarter-saving
β	Annual discount factor	0.967	0.967	0.968
χ	Owner housing-service premium	1.097	1.068	1.091
ξ	First-birth fixed utility cost	0.352	0.385	0.397
κ_1	First-birth taste dispersion	0.124	0.123	0.128
κ_C	Later-birth taste dispersion	0.363	0.350	0.348
θ_0	Bequest weight	0.106	0.106	0.109
γ	Child-benefit curvature	0.101	0.101	0.098
κ_T	Tenure-choice dispersion	0.014	0.014	0.015
ψ	Child-benefit level	0.178	0.172	0.174
h_P	Basic space requirement for parents (rooms)	2.504	2.562	2.492
H_0	Housing supply coefficient (derived)	6.757	7.546	7.289

H_0 is derived at normalized population $N_0 = 1$. Fixed inputs: $\sigma = 2$; $\alpha_0 = 0.733$; annual interest 2%; financed share $\phi = 0.8$.

Calibration comparison

- The soft rule fits best. Targets and weights are the same across specifications.
- Mean rooms: 5.957 (soft), 6.241 (hard), 6.131 (quarter-saving), against a target of 5.729.
- First-birth rooms difference: 1.324, 1.063 and 1.167, respectively, against 1.465.
- Children at age 25: 0.528, 0.524 and 0.521, against 0.810. Early fertility remains low in all three.
- Each rule is recalibrated. Hard and quarter-saving also require nonnegative net wealth after purchase.